Byeongjee Kang

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Research Interests

My research aims to automatically generate safe and efficient programs with provable guarantees. My recent interests are in adapting program execution to changing resource availability, through mechanisms such as checkpointing, restoration, and migration, with work across language design, compilers, and runtime systems. My work builds on LLVM, MLIR, Rust, and WebAssembly. I am also broadly interested in parallel programming, concurrency testing, and formal verification.

Technical Skills

Areas: Compiler engineering, static analysis, embedded systems, concurrency testing, formal verification

Languages & Tools: Rust, C++, Python, SML, Maude, Futhark, WebAssembly, LLVM, MLIR, rustc (MIR)

Education

PhD — **Carnegie Mellon University**, Software Engineering
Advised by Prof. Limin Jia
Pittsburgh, USA
Aug 2023 – May 2028 (expected)

BS — **Pohang University of Science and Technology (POSTECH)**, Computer Science and Engineering & Mathematics (double major)
Pohang, Korea
Feb 2018 – Feb 2023

- GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023)
- Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA
- Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)

Work Experiences

Microsoft Research, Research intern at Data Systems Group mentored by Kaushik Rajan
Redmond, WA, USA
May 2025 – Aug 2025

- Built auto-vectorizing compiler optimization passes using MLIR and Polygeist, vectorizing loops with control-flow dependencies that are not supported by existing loop vectorizers

Kodebox, Software engineer as military service alternative
Seoul, Korea
Jan 2020 – Dec 2021

- Developed the consensus engine of CodeChain Foundry, an open-source blockchain engine with a composable module system, implementing BLS signature aggregation for cryptographic vote verification and the interface between modules and the consensus engine
- Built Zuzu, a web-based cap table management software, as a full-stack engineer

Publications

Bao: Automatic Region Placement and Memory Allocation for Intermittent Computing
LLVM-based compiler that optimizes checkpoint placement and memory allocation for intermittently powered devices by solving a mixed-integer linear program
Byeongjee Kang, Jan Hoffmann, Limin Jia, Brandon Lucia, Feras Saad
(International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2027 (to appear)) [code]

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction Aug 2026
MLIR-based compiler from high-level languages to WebAssembly that takes advantage of high-level Wasm features such as stack switching
Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia
10.1145/3828170.3828172 (International Conference on Managed Programming Languages and Runtimes (MPLR) 2026) [code]

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM Oct 2025
Concurrency testing platform for Java. Formalized a subset of Java concurrency semantics and proved that scheduling only at synchronization points is complete for data-race-free programs
Ao Li, **Byeongjee Kang**, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye
10.1145/3764119 (Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) 2025) [code]

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits Sept 2024
Sam Westrick, Pengyu Liu, **Byeongjee Kang**, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar
ieeexplore.ieee.org/document/10821338 (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper))

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems July 2024
Refutation-complete symbolic model checking of distributed systems in Maude using narrowing with heuristic search
Byeongjee Kang, Kyungmin Bae
10.1016/j.scico.2024.103097 (Science of Computer Programming 2024) [code]

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search Dec 2022
Byeongjee Kang, Kyungmin Bae
dl.acm.org/doi/10.1145/3563822.3568017 (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022) [code]

Software

- **Bao**: LLVM toolchain that instruments programs for batteryless devices by solving a mixed-integer linear program
- **WAMI**: MLIR-based compilation pipeline for WebAssembly
- **OCaMI** (work in progress): Compiler from a functional subset of OCaml to WebAssembly through WAMI
- **rustlane** (work in progress): ISPC-style SPMD programming for Rust, lowering scalar control flow to masked SIMD
- **Fray**: Controlled concurrency testing framework for the JVM
- **heuristic-narrowing-search**: Maude implementation of narrowing-based symbolic reachability analysis
- **Simperby**: BFT blockchain engine for decentralized autonomous organizations (DAOs). Designed Vetomint, its consensus protocol, which extends Tendermint with leader veto and early termination
- **CodeChain Foundry**: Open-source blockchain engine based on composable module system

Ongoing Research Projects

- **Language-Level Program Migration**: Compiler and runtime support for migrating live Rust programs under strict time and energy budgets by erasing reinitializable state at the source and rebuilding it at the destination
- **Probabilistic Energy Estimation**: Probabilistic modeling and analysis tools for estimating energy consumption of programs on resource-constrained devices
- **Cumulative Memory Analysis**: Type-based automatic amortized resource analysis that infers symbolic bounds on memory usage integrated over execution time

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation) Aug 2023 – July 2028
- Received total of USD 120,000 during graduate studies

Global Leadership Program (CSE Department, POSTECH) Sept 2019 – Feb 2023
- Scholarship and research support program for selected students in the CS department

Domun Scholarship (POSTECH) Feb 2019
- Scholarship for freshmen with the highest GPA

Program for Highly Dedicated Students (POSTECH) Feb 2018 – Feb 2023
- Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year

Presidential Science Scholarship (Ministry of Education of Korea) Feb 2018 – Feb 2023
- National scholarship offered to top freshmen students in the field of science and technology

Talks

- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [recording]

Service

- Artifact Evaluation Committee, POPL 2027